

Department of Mathematics, IIT Madras
MA2031-Linear Algebra for Engineers
Assignment-III

1. Show that each of the maps given below is not an inner product on the given vector space.
 - (a) $\langle x, y \rangle = ac$ for $x = (a, b)$, $y = (c, d)$ in $\mathbb{R}^2$.
 - (b) $\langle x, y \rangle = ac$ for $x = (a, b)$, $y = (c, d)$ in $\mathbb{C}^2$.
 - (c) $\langle p, q \rangle = \int_0^1 p'(t)q(t) dt$ for $p, q \in \mathbb{R}[t]$.
 - (d) $\langle x, y \rangle = \int_0^1 x'(t)y'(t) dt$ for $x, y \in C^1[0, 1]$.
 - (e) $\langle f, g \rangle = \int_0^{1/2} f(t)g(t) dt$ for $f, g \in C[0, 1]$.
2. Let $A = [a_{ij}] \in \mathbb{R}^{2 \times 2}$. For $x, y \in \mathbb{R}^2$, let $f_A(x, y) = y^t A x$. Show that f_A is an inner product on $\mathbb{R}^2$ iff $a_{12} = a_{21}$, $a_{11} > 0$, $a_{22} > 0$, and $a_{11}a_{22} - a_{12}a_{21} > 0$.
3. Let B be a basis for a finite dimensional ips V . Let $y \in V$ be such that $\langle x, y \rangle = 0$ for all $x \in B$. Show that $y = 0$.
4. Let V be an inner product space, and let $x, y \in V$. Show the following:
 - (a) $\|x + \alpha y\| = \|x - \alpha y\|$ for all $\alpha \in \mathbb{F}$ iff $\langle x, y \rangle = 0$.
 - (b) If $\|x + y\| = \|x\| + \|y\|$, then $\{x, y\}$ is a linearly dependent set.
5. Let V be a complex ips. Show that $\operatorname{Re}\langle ix, y \rangle = -\operatorname{Im}\langle x, y \rangle$ for all $x, y \in V$.
6. (Polarization Identity): Let V be an ips over $\mathbb{F}$. Let $x, y \in V$. Show the following:
 - (a) If $\mathbb{F} = \mathbb{R}$, then $4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$.
 - (b) If $\mathbb{F} = \mathbb{C}$, then $4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2$.
7. For $1 \leq j \leq n$, let $a_j \geq 0$ and $b_j \geq 0$. Show that $(\sum_{j=1}^n a_j b_j)^2 \leq (\sum_{j=1}^n (j a_j^2)) (\sum_{j=1}^n (a_j / j^2))$.
8. Let $W = \{x \in \mathbb{R}^4 : x \perp (1, 0, -1, 1), x \perp (2, 3, -1, 2)\}$, where $\mathbb{R}^4$ is the real ips with standard inner product. Show that W is a subspace of $\mathbb{R}^4$. Also, find a basis for W .
9. For vectors x, y in a real ips, prove that $x + y \perp x - y$ iff $\|x\| = \|y\|$.
10. Consider the standard basis for $\mathbb{R}^n$. Take $x = (a_1, \dots, a_n)$. Verify Fourier expansion and Parseval's identity.
11. Take $x = (n, n-1, \dots, 1) \in \mathbb{R}^n$. Consider the orthonormal set $\{e^2, \dots, e^{n-1}\}$. Verify Bessel's inequality.
12. Let $\{u_1, u_2, \dots, u_k\}$ be an orthogonal set in a real ips V . Let $a_1, \dots, a_k \in \mathbb{R}$. Is it true that $\|\sum_{i=1}^k a_i u_i\|^2 = \sum_{i=1}^k |a_i|^2 \|u_i\|^2$?
13. For a subset S of an ips V , define $S^\perp = \{x \in V : \langle x, u \rangle = 0, \text{ for all } u \in S\}$. Show the following:
 - (a) $V^\perp = \{0\}$ (b) $\{0\}^\perp = V$ (c) $S^\perp$ is a subspace of V (d) $S \subseteq S^{\perp\perp}$.
14. Let W be a subspace of a finite dimensional ips V . Show the following:
 - (a) $V = W + W^\perp$ (b) $W \cap W^\perp = \{0\}$ (c) $W^{\perp\perp} = W$.
15. Show that $\{\sin t, \sin(2t), \dots, \sin(mt)\}$ is linearly independent in $C[0, 2\pi]$.
16. Consider $\mathbb{R}^3$ with the standard inner product. Apply Gram-Schmidt process on the given set of vectors.
 - (a) $\{(1, 2, 0), (2, 1, 0), (1, 1, 1)\}$ (b) $\{(1, 1, 1), (1, -1, 1), (1, 1, -1)\}$

✓ 17. Consider $\mathbb{R}^3$ with the standard inner product. In each of the following, find a vector of norm 1 which is orthogonal to the given two vectors:

- ✓ (a) $(2, 1, 0), (1, 2, 1)$ ✓ (b) $(1, 2, 3), (2, 1, -2)$ ✓ (c) $(0, 2, -1), (-1, 2, -1)$.

✓ 18. Consider $\mathbb{R}^4$ with the standard inner product. In each of the following, find the set of all vectors orthogonal to both u and v .

- ✓ (a) $u = (1, 2, 0, 1), v = (2, 1, 0, -1)$ ✓ (b) $u = (1, 1, 1, 0), v = (1, -1, 1, 1)$

✓ 19. Consider $\mathbb{C}^3$ with the standard inner product. Find an orthonormal basis for the subspace spanned by the vectors $(1, 0, i)$ and $(2, 1, 1 + i)$.

✓ 20. Consider the polynomials $u^0(t) = 1, u^1(t) = t, u^2(t) = t^2$ in $\mathbb{R}_2[t]$. Using Gram-Schmidt orthogonalization, find orthogonal polynomials obtained from u^1, u^2, u^3 with respect to the inner product:

(a) $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$.

21. Find the best approximation of $v \in V$ from U in the following:

- (a) $V = \mathbb{R}^3, v = (1, 2, 1), U = \text{span}\{(3, 1, 2), (1, 0, 1)\}$.
(b) $V = \mathbb{R}^3, v = (1, -1, 1), U = \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : \alpha + \beta + \gamma = 0\}$.
(c) $V = \mathbb{R}^4, v = (1, 0, -1, 1), U = \text{span}\{(1, 0, -1, 1), (0, 0, 1, 1)\}$.
(d) $V = \mathbb{P}_3[t], v = t^3, U = \text{span}\{1, 1+t, 1+t^2\}, \langle p, q \rangle = \int_0^1 p(t)q(t) dt$.
(e) Find the QR decomposition of the following matrices:

(a) $A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ (b) $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$.
